EXPERIMENT 03

Morphological Analysis Using NLTK

Aim

To demonstrate basic morphological analysis by tokenizing words and identifying their stems using NLTK.

Procedure

1. Install NLTK using pip if required.
2. Import NLTK's PorterStemmer.
3. Prepare a list of words.
4. Apply stemming to each word.
5. Display each word with its stem.

Program

```
# Install first: pip install nltk
from nltk.stem import PorterStemmer

stemmer = PorterStemmer()
words = ["playing", "played", "plays", "studies", "happily"]

for word in words:
    print(word, "->", stemmer.stem(word))
```

Output

```
OUTPUT — EXPERIMENT 03
playing -> play
played -> play
plays -> play
studies -> studi
happily -> happili
```

Result

Thus, NLTK was used to demonstrate a basic form of morphological analysis through stemming.